

What is Customer Churn Prediction?

Anticipate Departures: Forecasts if a customer will discontinue using a product or service.

Leverage Data: Utilizes historical customer data and behavioral patterns to identify at-risk individuals.

Strategic Intervention: Facilitates targeted retention strategies before customers decide to leave.



Key Data Features Driving Churn Prediction



Demographics

Age, Gender, family status, SeniorCitizen



Usage Patterns

Frequency, duration, feature engagement



Interactions

TechSupport, OnlineBackup, DeviceProtection



Subscription History

Payment history, contract type, plan changes, MonthlyCharges



Popular Machine Learning Models for Churn Prediction

Decision Tree

Simple and easy to interpret, deal for binary classification problems like Customer Churn (Yes / No)

Random Forest

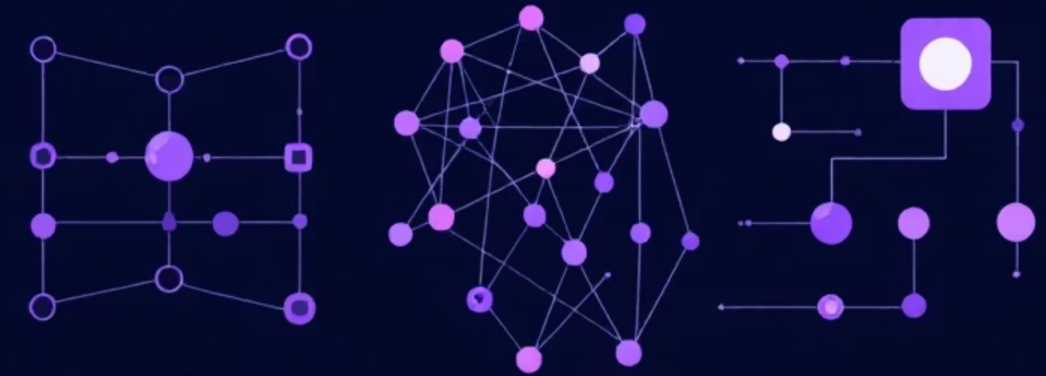
Handles complex data and smaller datasets effectively.

Gradient Boosting Machines

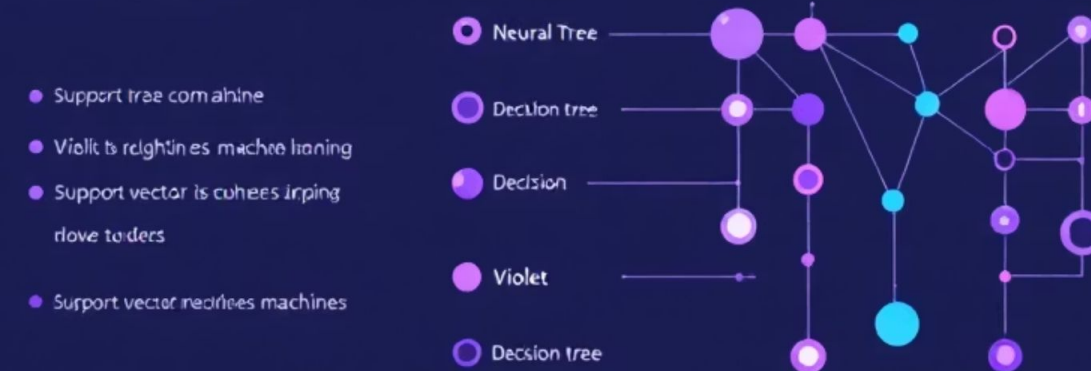
High accuracy, robust with feature interactions.

Model choice depends on data size, complexity, and interpretability needs.

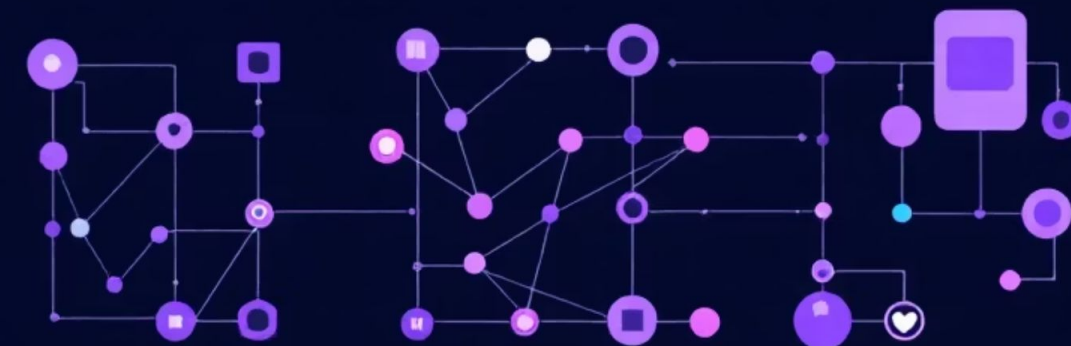
Neural Network



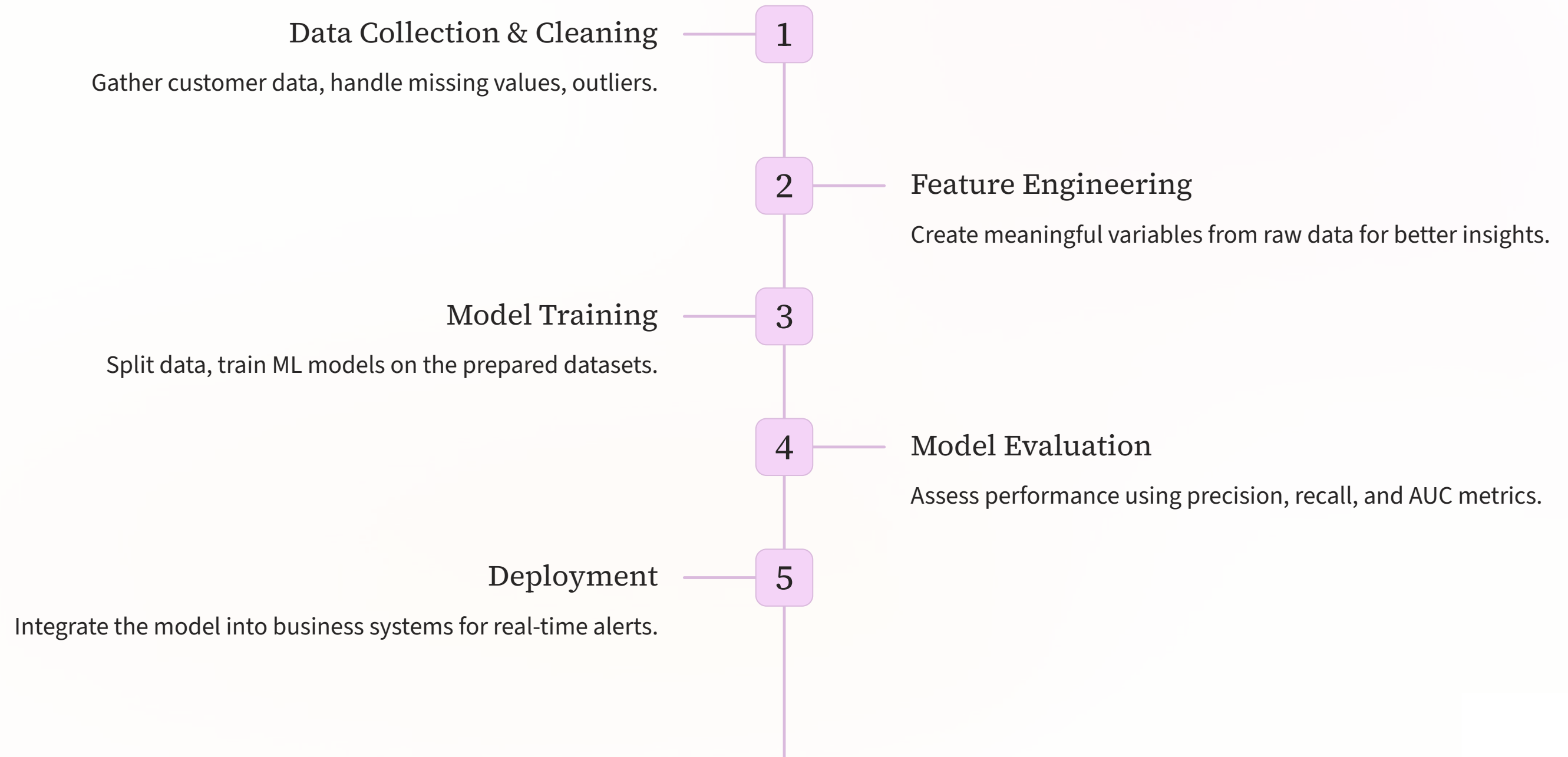
Decision Tree Info



Support Vector Machine



Step-by-Step ML Project Workflow



Real-World Example: Telecom Churn Prediction

Dataset: Over 7,000 customers with extensive usage, billing, and support interaction data.

Model: Utilized a Random Forest model for robust prediction.

Performance: Achieved 85% recall and 80% precision, proving strong predictive power.

Outcome: Targeted retention campaigns led to a 15% reduction in churn within six months, demonstrating clear ROI.



Thank You

We appreciate your time and attention.

